12/04/2024

Phase-I
CODE-A

Aakash
Medical | IIT-JEE | Foundations

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# FINAL TEST SERIES for NEET-2024

## Test-7

MM : 720

Time : 3 Hrs. 20 Mins.

**Topics covered :**

| | | |
|---|---|---|
| ***Physics*** | : | Alternating Current, Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics |
| ***Chemistry*** | : | Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehyds, Ketones and Carboxylic Acids |
| ***Botany*** | : | Microbes in Human Welfare, Organisms and Populations |
| ***Zoology*** | : | Human Health and Disease |

**Instructions :**

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

## PHYSICS

**Choose the correct answer :**

### SECTION - A

1. The SI unit of power factor is

(1) watt
(2) J s
(3) ampere
(4) No unit

2. For an electromagnetic wave, the direction of propagation of wave is in the direction of

(1) $\vec{E}$

(2) $\vec{B}$

(3) $\vec{B} \times \vec{E}$

(4) $\vec{E} \times \vec{B}$

3. A D.C. ammeter shows 1 A reading for a circuit containing coil connected to 100 V D.C. source. The same coil carries 0.5 A current when connected to a 100 V, 50 Hz A.C. supply. The inductance of coil is nearly

(1) $\frac{\sqrt{3}}{\pi}$ H

(2) $\sqrt{5}\pi$ mH

(3) $\frac{\sqrt{3}}{\pi}$ mH

(4) $\frac{3}{\pi}$ mH

4. In a series LCR circuit, the capacitance is changed from C to $\frac{C}{4}$. For the resonating frequency to remain unchanged, the inductance should becomes

(1) 4 times (2) $\frac{1}{4}$ times

(3) 2 times (4) $\frac{1}{2}$ times

5. Which of the following statement is correct for a transformer?

(1) A step up transformer increases the power

(2) A step down transformer decreases the current

(3) Using laminated core in transformer reduces energy loss

(4) If transformation ratio, $K$ = 4 then output power is four times the input power

6. Electric field between the plates of a capacitor varies as $E = 4t^2$. The displacement current between the plates is proportional to

(1) $t^2$ (2) $t$

(3) $t^3$ (4) $\frac{1}{t^2}$

7. A plane electromagnetic wave propagates in vacuum where amplitude of electric field is 30 V/m. The amplitude of magnetic field will be

(1) $10^{-7}$ T

(2) $\sqrt{2}\times10^{-7}$ T

(3) $10^{-9}$ T

(4) $\sqrt{2}\times10^{-9}$ T

8. The electromagnetic rays which is widely used to disinfect water is

(1) UV Rays (2) IR Rays

(3) X-Rays (4) $\gamma$-Rays

9. Light travelling in air strikes an air-glass refracting interface at angle '$\theta$'. If the refractive index of glass is 1.5 then it will

(1) Undergo total internal reflection for $\theta$ = 30°

(2) Undergo total internal reflection for $\theta$ = 45°

(3) Undergo total internal reflection for $\theta$ = 60°

(4) Not undergo total internal reflection for any value of '$\theta$'

10. A giant refracting telescope at an observatory has an objective lens of focal length 15 m. If an eye-piece of focal length 1.0 cm is used, then the angular magnification of the telescope at normal adjustment will be

(1) 1500

(2) 150

(3) 15

(4) 1.5

11. A small bulb is placed at the bottom of tank containing water to a depth of 100 cm. The area of the surface of water through which light from the bulb can emerge out is approximately

[Refractive index of water is 1.33 (Consider the bulb to be a point source)]

(1) 12 $m^2$

(2) 8 $m^2$

(3) 6.5 $m^2$

(4) 4 $m^2$

12. The critical angle for light going from medium $x$ into medium $y$ is $\theta$. The speed of light in medium x is $v$. The speed of light in medium $y$ is

(1) $v(1 - \cos\theta)$

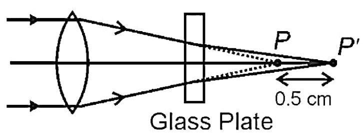
(2) $\frac{v}{\sin\theta}$

(3) $\frac{v}{\cos\theta}$

(4) $v\cos\theta$

Space for Rough Work

13. Rays from a lens are converging towards a point $P$, as shown in figure. The thickness of glass plate of refractive index 2 kept between the lens and point $P$, so that image will be formed at $P'$, will be



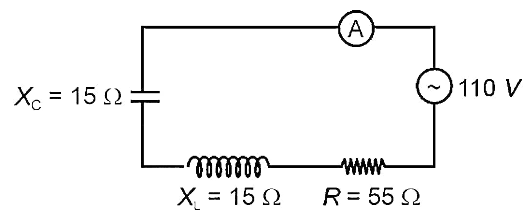

(1) 1 cm (2) 3.2 cm
(3) 5 cm (4) 2.4 cm

14. If a transparent parallel plate of uniform thickness $t$ having refractive index $\mu$ is interposed perpendicularly in the path of a light beam, the optical path is
(1) Increased by $(\mu - 1)t$
(2) Decreased by $\mu$
(3) Decreased by $(\mu - 1)t$
(4) Increased by $\mu$

15. **Assertion (A):** For best contrast between maxima and minima in the interference pattern in YDSE, the intensity of light waves emerging out of the two slits should be equal.
**Reason (R):** The intensity of interference pattern is proportional to square of amplitude of resultant wave.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true and the reason is not the correct explanation of the assertion
(3) Assertion is true but Reason is false
(4) Both Assertion and Reason are false

16. A ray of unpolarised light is incident on a glass plate at polarising angle 57°. Then choose the correct statement.
(1) The reflected ray and the transmitted ray both will be completely polarised
(2) The reflected ray will be completely polarised and the transmitted ray will be partially polarised
(3) The reflected ray will be partially polarised and the transmitted ray will be completely polarised
(4) The reflected and transmitted both rays will be partially polarised

17. The reading of ammeter in the circuit shown will be




(1) 2 A (2) 4 A
(3) 3 A (4) 5 A

18. What will be the angular width of central maxima in Fraunhofer diffraction when the light of wavelength 5000 Å is used and slit width is $10 \times 10^{-5}$ cm?
(1) 2 rad (2) 3 rad
(3) 1 rad (4) 4 rad

19. In an LCR series circuit, $R = 10\ \Omega$, $X_L = 20\ \Omega$ and $X_C = 10\ \Omega$. The average value of current for a complete cycle, for source voltage $V = 20\sin(30t)$ is
(1) 2 A (2) 0.5 A
(3) 3 A (4) Zero

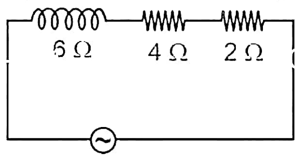
20. Out of the following, choose the ray which does not travel with the speed of light in vacuum
(1) X-rays (2) Microwaves
(3) $\gamma$-rays (4) $\beta$-rays

**Space for Rough Work**

21. Which among the following has the largest wavelength?
    (1) Radio wave (2) X-ray
    (3) Ultraviolet ray (4) Infra-red ray

22. An electromagnetic wave of intensity 30 W/m$^2$ is incident on a surface of area 2 m$^2$. If the surface is perfectly absorbing, then the amount of momentum transferred to the surface in 10 s will be
    (1) $2 \times 10^{-6}$ kg m/s (2) $2 \times 10^{6}$ kg m/s
    (3) $4 \times 10^{-9}$ kg m/s (4) $10^{-8}$ kg m/s

23. The net impedance of the following circuit is



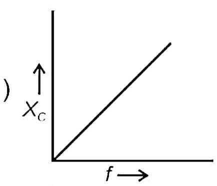

    (1) $6\sqrt{2}\ \Omega$ (2) $2\ \Omega$
    (3) $4\ \Omega$ (4) $6\ \Omega$

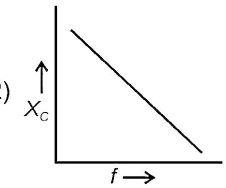
24. The r.m.s. value of current from $t = 0$ to $t = 3$ s for the current $i = (3t)$ A is
    (1) 2 A (2) 1.45 A
    (3) 4 A (4) 5.19 A

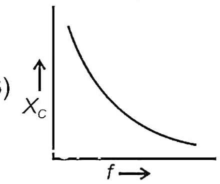
25. Consider the following two statements regarding transformers and choose the correct one.
    **Statement-A :** For step up transformer, $V_s > V_p$ so $N_s > N_p$ and $I_s < I_p$
    **Statement-B :** For step down transformer, $V_s < V_p$ so $N_s < N_p$ and $I_s > I_p$
    (Here symbols have their usual meaning)
    (1) Only statement A is correct
    (2) Only statement B is correct
    (3) Both statements are correct
    (4) Neither statement A nor statement B is correct

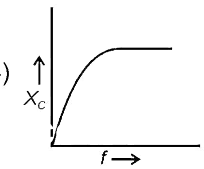
26. An ideal transformer converts a 20 V input voltage in primary coil to 100 V output in secondary coil. The current in secondary coil if the current in primary coil is 20 A is
    (1) 4 A (2) 20 A
    (3) 0.3 A (4) 40 A

27. Which of the following curves correctly represents the variation of capacitive reactance ($X_C$) with frequency $f$?

(1) 


(2) 


(3) 


(4) 


28. There is a current of 20 mA passing through an ideal inductor when connected to a 100 V, 20 Hz supply. The power consumed is
    (1) 1 W (2) 25 W
    (3) 39 W (4) Zero

29. A small insect is at 60 cm from a concave mirror of focal length 20 cm. What is the image distance from the mirror?
    (1) 30 cm (2) 20 cm
    (3) 15 cm (4) 18 cm

30. The minimum distance between the real object and its real image for concave mirror is
    (1) $f$ (2) $2f$
    (3) $4f$ (4) Zero

31. A compound microscope has two lenses. The magnifying power of one is 8 and the combined magnifying power is 96. The magnifying power of the other lens is
    (1) 8 (2) 12
    (3) 14 (4) 6

32. What is angular width between central maxima and first ordered maxima of diffraction pattern due to single slit of width 0.25 mm, when light of wavelength 600 nm is incident on it normally?
    (1) $3.6 \times 10^{-3}$ rad (2) $4.2 \times 10^{-3}$ rad
    (3) $1.2 \times 10^{-3}$ rad (4) $2.4 \times 10^{-3}$ rad

Space for Rough Work

33. Choose the correct statement among the following.

(1) Polarization of light supports wave nature while diffraction of light supports particle nature

(2) Polarization of light supports particle nature of light while diffraction of light supports wave nature

(3) Both polarization of light and diffraction supports wave nature of light

(4) Both polarization of light and diffraction support particle nature of light

34. Two polarising sheets are placed with their plane parallel so that intensity of transmitted light is maximum. The angle through which either sheet must be turned so that light intensity drops to half of maximum value is

(1) 30°

(2) 45°

(3) 135°

(4) Both (2) and (3) are correct

35. In single slit diffraction experiment, first minima of red light $\lambda_1$ = 660 nm coincides with first maxima of another wavelength $\lambda_2$. The value of $\lambda_2$ is

(1) 440 nm (2) 470 nm

(3) 550 nm (4) 690 nm

**SECTION-B**

36. A prism is made of glass of refractive index $\sqrt{3}$. The refracting angle of prism is $A$. If angle of minimum deviation is equal to $A$, then the value of $A$ is

(1) 30° (2) 45°

(3) 60° (4) 90°

37. Dispersion of light in a medium implies that

(1) Light of different wavelengths travel with equal speed in medium

(2) Light of different frequencies travel with equal speed in medium

(3) Refractive index of medium is different for different wavelengths

(4) All of above are correct

38. When a ray of light enters a glass slab from air

(1) Its speed increases

(2) Its frequency decreases

(3) Its wavelength decreases

(4) Neither wavelength nor frequency change

39. Two coherent monochromatic light waves of amplitude $3A$ and $2A$ interfere at a point on screen having a phase difference of $\frac{\pi}{3}$ rad. The resultant intensity at that point will be proportional to

(1) $5A^2$ (2) $13A^2$

(3) $19A^2$ (4) $6A^2$

40. The velocity of light in air is $3 \times 10^8$ m s$^{-1}$ and in water $2.25 \times 10^8$ m/s. The polarising angle for air water interface will be

(1) 15° (2) 30°

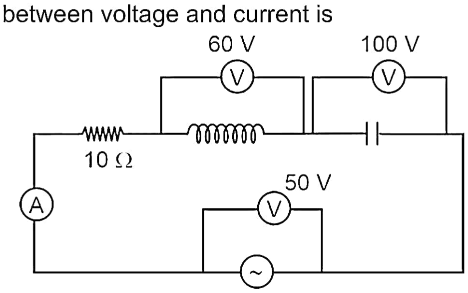
(3) 53° (4) 37°

41. The deviation $\delta$ of a ray produced by a thin prism of small angle $A$ is

(1) $\frac{\mu - 1}{A}$ (2) $\frac{A}{(\mu - 1)}$

(3) $(\mu - 1)A$ (4) $(\mu^2 - 1)A$

42. In the circuit shown below the phase difference between voltage and current is




(1) 45° (2) 53°

(3) 37° (4) 30°

Space for Rough Work

43. Consider the following two statements.

**Statement-A :** A hot wire ammeter reads the RMS value of current.

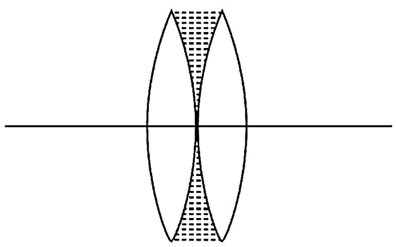
**Statement-B :** A hot wire ammeter can read AC but cannot read DC.

The correct statement(s) is/are

(1) Only A (2) Only B

(3) Both A and B (4) Neither A nor B

44. Two identical thin equiconvex lenses made of glass ($\mu$ = 1.5) are placed in contact. Their combine power is *P*. If the space between them is filled with water $\mu$ = 1.33 as shown then power of the combination



(1) Will be greater than *P*

(2) Will be less than *P*

(3) May be less than or greater than *P*

(4) Will becomes zero

45. An equiconvex lens has focal length $\frac{80}{3}$ cm and radius of curvature 40 cm. If one face of the lens is silvered then the effective focal length of the mirror so formed will be

(1) $-\frac{80}{3}$ cm (2) – 8 cm

(3) – 40 cm (4) $-\frac{40}{3}$ cm

46. An electromagnetic wave of intensity 100 W/m$^2$ falls on a perfectly reflecting surface. The force exerted by it on a 6 cm$^2$ portion of surface will be

(1) $4 \times 10^{-10}$ N (2) $10^{-12}$ N

(3) $4 \times 10^{-6}$ N (4) $0.66 \times 10^{-8}$ N

47. A series LCR circuit has a inductive reactance of 100 $\Omega$, a capacitive reactance of 200 $\Omega$ and resistor of 100 $\Omega$. It is connected to an A.C. source of $150\sqrt{2}$ V, $\frac{500}{\pi}$ Hz. The power dissipated in the resistor will be

(1) 450 W (2) 225 W

(3) 311.5 W (4) 100 W

48. Column-I contains quantities while column-II contains their corresponding SI units. Match the columns and choose the correct option.

| | Column-I | | Column-II |
|---|---|---|---|
| A. | Impedance | (P) | Ampere |
| B. | Wattless current | (Q) | Ohm |
| C. | Quality factor | (R) | Volt |
| D. | RMS voltage | (S) | Unitless |

(1) A $\to$ Q; B $\to$ P; C $\to$ S; D $\to$ R

(2) A $\to$ P; B $\to$ Q; C $\to$ R; D $\to$ S

(3) A $\to$ Q; B $\to$ R; C $\to$ P; D $\to$ S

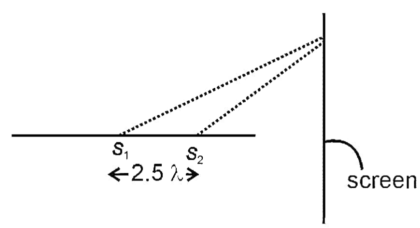
(4) A $\to$ S; B $\to$ Q; C $\to$ R; D $\to$ P

49. Monochromatic light of wavelength 8000 Å is used in YDSE, if fifth bright fringe is observed at 2 cm from centre then third dark fringe will be at

(1) 1 cm (2) 2 cm

(3) 4 cm (4) 0.5 m

50. In the following modified set-up of YDSE, the number of minima formed on screen is




(1) 5 (2) 7

(3) 3 (4) 4

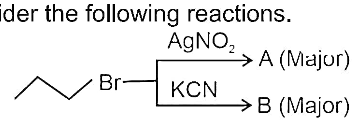
Space for Rough Work

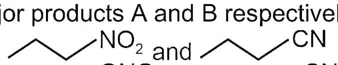
# CHEMISTRY

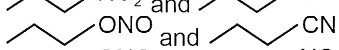
## SECTION - A

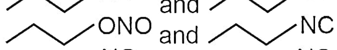
51. Which among the following is an incorrect statement?
    (1) $S_N1$ reaction is accelerated in polar protic solvent.
    (2) In $S_N2$ reaction, transition state is formed.
    (3) Complete retention in configuration takes place in $S_N1$ reaction.
    (4) Inversion in configuration takes place in $S_N2$ reaction.

52. Consider the following reactions.




Major products A and B respectively are
(1) 

(2) 

(3) 

(4) 


53. Consider the following reaction

$$CH_3-\underset{|}{\overset{Ph}{CH}}-CH=CH_2 \xrightarrow[\text{(II) } H_2O_2/OH^-]{\text{(I) } B_2H_6} P \text{ (Major)}$$

Major product P is

(1) $CH_3-\overset{Ph}{\overset{|}{CH}}-CH_2-CH_2-OH$

(2) $CH_3-\overset{Ph}{\overset{|}{CH}}-\underset{OH}{\underset{|}{CH}}-CH_3$

(3) $CH_3-\overset{Ph}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_2-CH_3$

(4) $\underset{OH}{\underset{|}{CH_2}}-\overset{Ph}{\overset{|}{CH}}-CH_2-CH_3$

54. Given below are two Statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

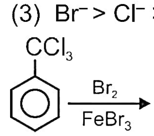
**Assertion (A):** Benzyl halide is an aryl halide.

**Reason (R):** In benzyl halide, halogen atom is bonded to an $sp^3$-hybridised carbon atom attached to an aromatic ring.

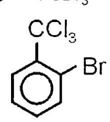
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) (A) is correct but (R) is incorrect
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

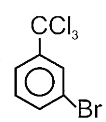
55. Total number of isomeric products (including stereoisomers) obtained when 2-chlorobutane is heated with alcoholic KOH is
(1) One (2) Two
(3) Three (4) Four

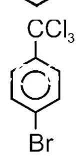
56. The increasing order of nucleophilicity in polar aprotic medium would be
(1) $F^- > Cl^- > Br^- > I^-$ (2) $I^- > Br^- > Cl^- > F^-$
(3) $Br^- > Cl^- > I^- > F^-$ (4) $I^- > F^- > Cl^- > Br^-$

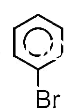
57. 

P (Major), Major product P is

(1) 

(2) 

(3) 

(4) 


**Space for Rough Work**

58. Given below are two statements

**Statement-I:** The acidic strength of alcohols decreases in the order: Primary > Secondary > Tertiary.

**Statement-II:** Phenols do not react with aqueous NaOH.

In the light of the above statements, choose the **correct** answer from the options given below.

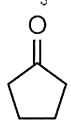
(1) Statement-I is true but statement-II is false
(2) Statement-I is false but statement-II is true
(3) Both statement-I and statement-II are false
(4) Both statement-I and statement-II are true

59. Which of the following reactions will give primary alcohol as major product?

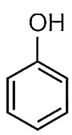
(1)  $\xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3MgBr}$

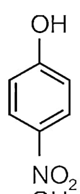
(2) $CH_3CHO \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3CH_2MgBr}$

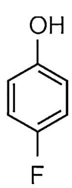
(3) $CH_3-\overset{\overset{O}{||}}{C}-C_2H_5 \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3MgBr \text{ (excess)}}$

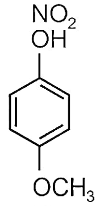
(4)  $\xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3MgBr}$

60. Compound having lowest $pK_a$ value among the following is

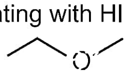
(1) 

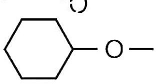

(2) 

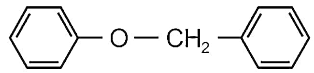

(3) 

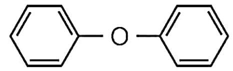

(4) 


61. The ether which is most difficult to be cleaved by heating with HI among the following is

(1) 

(2) 

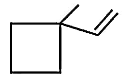
(3) 

(4) 

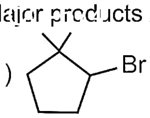
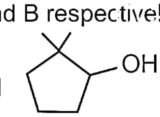
62. Williamson synthesis mostly involves

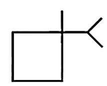
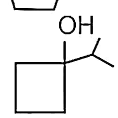
(1) $S_N1$ attack of an alkoxide ion on 3°-alkyl halide
(2) $S_N1$ attack of an alkoxide ion on 2°-alkyl halide
(3) $S_N2$ attack of an alkoxide ion on 1°-alkyl halide
(4) $S_N2$ attack of an alkoxide ion on 3°-alkyl halide

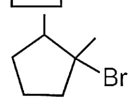
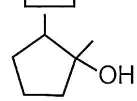
63. Consider the following reaction

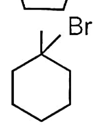
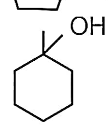
 $\xrightarrow{HBr}$ A (Major) $\xrightarrow[\text{NaOH}]{\text{Aqueous}}$ B (Major)

Major products A and B respectively are

(1)  and 

(2)  and 

(3)  and 

(4)  and 

64. Given below are two Statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A):** Addition of HCN to $CH_3CHO$ gives an optically active compound.

**Reason (R):** Nucleophilic addition of HCN in aldehyde with two or more carbon gives a chiral product.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct
(4) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

**Space for Rough Work**

65. $C_2H_5-C\equiv CH \xrightarrow[\text{(ii) } CH_3Br]{\text{(i) } CH_3MgBr}$ Product

The product is

(1) $CH_3-(CH_2)_3-CH_3$

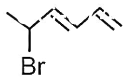
(2) $C_2H_5-\underset{CH_3}{\underset{|}{C}}=\underset{Br}{\underset{|}{CH}}$

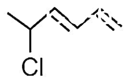
(3) $C_2H_5CH=CH-CH_3$

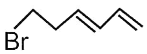
(4) $C_2H_5-C\equiv C-CH_3$

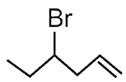
66. Major product of the given reaction is

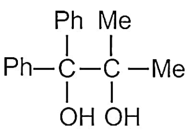
 $\xrightarrow[CCl_4]{\text{NBS, h}\nu}$ Product

(1) 

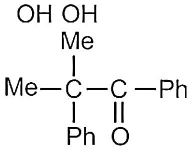

(2) 

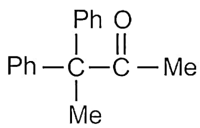

(3) 

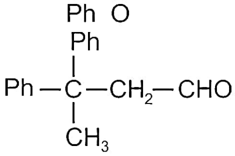

(4) 

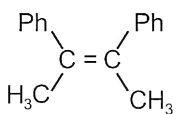

67. 

$\xrightarrow{H_2SO_4}$ P(Major), P is

(1) 


(2) 


(3) 

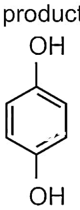

(4) 

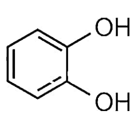

68. Which of the following is a polar aprotic solvent?

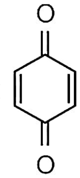
(1) $H_2O$ (2) DMSO

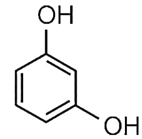
(3) $C_2H_5OH$ (4) $CS_2$

69. When phenol is treated with chromic acid then the product formed is

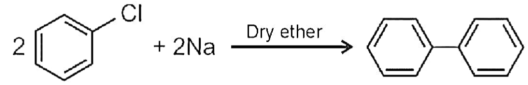
(1) 


(2) 


(3) 


(4) 

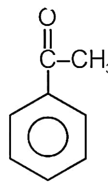

70. The given reaction is called




(1) Wurtz reaction (2) Fittig reaction

(3) Wurtz-Fittig reaction (4) Swarts reaction

71. The reactivity order towards the nucleophilic addition reaction among the following compounds is

(I) 


(II) $CH_3-\overset{O}{\overset{||}{C}}-CH_3$

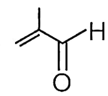
(III) $CH_3-\overset{O}{\overset{||}{C}}-H$

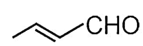
(1) I > II > III (2) II > III > I

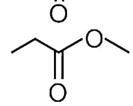
(3) III > II > I (4) I > III > II

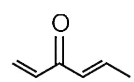
72. In the following reaction, the product [A] is

$CH_3CHO \xrightarrow[\Delta]{\text{dil.NaOH}} [A]$

(1) 


(2) 


(3) 


(4) 


73. In which of the following reaction hydrocarbon is produced?

(1) $CH_3COCH_2COOH \xrightarrow{\Delta}$

(2) $CH_3COONa + H_2O \xrightarrow{\text{Electrolysis}}$

(3) $CH_3COOC_2H_5 + NaOH \xrightarrow{\Delta}$

(4) $C_2H_5COOH + NaHCO_3 \xrightarrow{\Delta}$

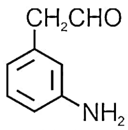
**Space for Rough Work**

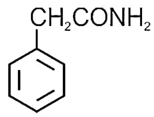
74. Compound 'A' undergoes formation of cyanohydrin which on hydrolysis gives lactic acid [$CH_3CH(OH)COOH$] therefore, compound 'A' is

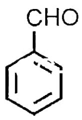
(1) Formaldehyde (2) Acetaldehyde

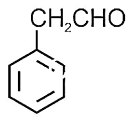
(3) Acetone (4) Benzophenone

75. The compound which will not form oxime with $NH_2OH$ is

(1) 


(2) 


(3) 


(4) 


76. The major product (A) in the following reaction is

$$R-Mg-X+CO_2 \xrightarrow{\text{Dry ether}} \xrightarrow{H_3O^+} A$$

(1) An aldehyde (2) A ketone

(3) An ether (4) A carboxylic acid

77 $F_3CCOOH$ (a) $HCOOH$ (b) $C_6H_5COOH$ (c)

The order of $pK_a$ of given acids is

(1) a > c > b (2) c > b > a

(3) b > c > a (4) a > b > c

78. Least reactive acid derivative towards hydrolysis is

(1) Ester (2) Acid amide

(3) Acid halide (4) Acid anhydride

79. $2D-\underset{\substack{|}}{\overset{D}{C}}=O + OH^- \longrightarrow X + Y$

X and Y will be

(1) $D-\overset{O}{\overset{||}{C}}-O^{\ominus}$, $D-C(D)_2-OH$

(2) $D-\overset{O}{\overset{||}{C}}-O^{\ominus}$, $D-C(D)(H)-OH$

(3) $H-\overset{O}{\overset{||}{C}}-O^{\ominus}$, $D-C(D)_2-OH$

(4) $H-\overset{O}{\overset{||}{C}}-O^{\ominus}$, $D-C(D)(H)-OH$

80. $C_2H_5COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) Red P}/X_2}$ Product, the **incorrect** statement for the reaction is

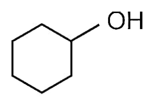
(1) The product formed is a chiral compound

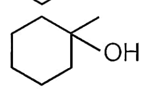
(2) This is Hell-Volhard-Zelinsky reaction

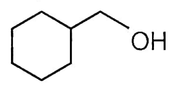
(3) $\alpha$ - halo acid is formed as major product

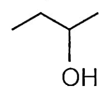
(4) Acyl halide is formed as major product

81. Which alcohol will react slowest with concentrated HCl and $ZnCl_2$?

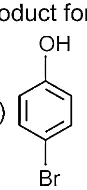
(1) 

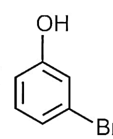

(2) 

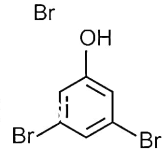

(3) 

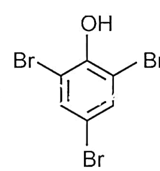

(4) 


82. When phenol reacts with $Br_2$/water, the major product formed is

(1) 


(2) 


(3) 


(4) 


Space for Rough Work

83. Given below are two statements

**Statement-I:** $(CH_3)_3CCl$ gives faster $S_N1$ reaction than $C_6H_5CH_2Cl$.

**Statement-II:** $C_6H_5\overset{\oplus}{C}H_2$ is less stable than $(CH_3)_3\overset{\oplus}{C}$.

In the light of the above statements, choose the **correct** answer from the options given below.

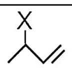
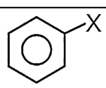
(1) Both statement-I and statement-II are true

(2) Both statement-I and statement-II are false

(3) Statement-I is true but statement-II is false

(4) Statement-I is false but statement-II is true

84. Match the column 'A' with column 'B'.

| | Column A | | Column B |
|---|---|---|---|
| (i) |  | a. | Aryl halide |
| (ii) |  | b. | Vinyl halide |
| (iii) |  | c. | Allyl halide |
| (iv) |  | d. | Alkyl halide |

Choose the **correct** option.

(1) (i)$\rightarrow$a, (ii)$\rightarrow$c, (iii)$\rightarrow$b, (iv)$\rightarrow$d

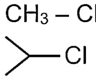
(2) (i)$\rightarrow$d, (ii)$\rightarrow$b, (iii)$\rightarrow$c, (iv)$\rightarrow$a

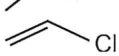
(3) (i)$\rightarrow$d, (ii)$\rightarrow$c, (iii)$\rightarrow$b, (iv)$\rightarrow$a

(4) (i)$\rightarrow$d, (ii)$\rightarrow$b, (iii)$\rightarrow$a, (iv)$\rightarrow$c

85. The decreasing order of rate of $S_N2$ reaction for the given compounds is

(I) $CH_3 - Cl$

(II) 

(III) 

(1) (III) > (II) > (I)

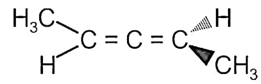
(2) (II) > (III) > (I)

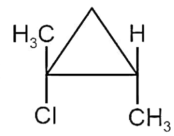
(3) (II) > (I) > (III)

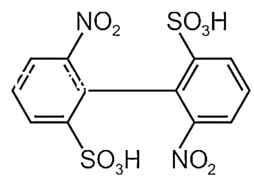
(4) (I) > (II) > (III)

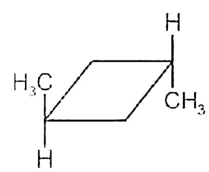
**SECTION-B**

86. Which among the following is an optically inactive molecule?

(1) 

(2) 

(3) 

(4) 

87. When phenyl magnesium bromide reacts with t-butanol, the product would be

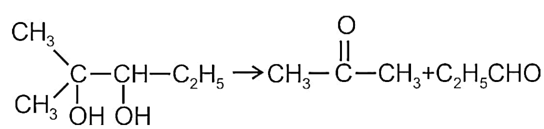
(1) Benzene

(2) Phenol

(3) t-butyl benzene

(4) t-Butyl phenyl ether

88. The given conversion can be achieved by



(1) Baeyer's reagent

(2) Dil. $H_2SO_4$

(3) $HIO_4$

(4) Acidified $K_2Cr_2O_7$

Space for Rough Work

89. Given below are two statements

**Statement-I:** 1-Chloropropane has higher boiling point than 2-Chloropropane.

**Statement-II:** For isomeric alkyl halides order of boiling point is 3° > 2° > 1°.

In the light of above statements choose the **correct** answer from options given below.

(1) Statement I is true and statement II is false

(2) Statement I is false and statement II is true

(3) Both statement I and statement II are false

(4) Both statement I and statement II are true

90. In which reaction, C–C bond formation does not take place?

(1) Reimer-Tiemann reaction

(2) Swarts reaction

(3) Fittig reaction

(4) Wurtz reaction

91. Given below are two statements

**Statement-I:** Reaction of methanal with $CH_3MgBr$ gives secondary alcohol as major product.

**Statement-II:** Reaction of Grignard reagent with carbonyl compound gives alcohol.

In the light of above statements choose the **correct** answer from options given below.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true and statement II is false

(4) Statement I is false and statement II is true

92. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A):** Ethanal and ethanol can be distinguished by iodoform reaction.

**Reason (R):** Aldehydes and ketones having '$CH_3-\overset{\overset{\large O}{||}}{C}-$' group give yellow precipitate in iodoform reaction.

In the light of above statements, choose the most appropriate answer from the options given below.

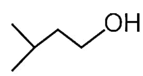
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

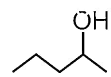
(2) (A) is correct but (R) is not correct

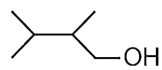
(3) (A) is not correct but (R) is correct

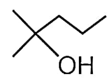
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

93. The compound which gives blue colouration in Victor Meyer's test is

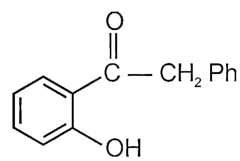
(1) 

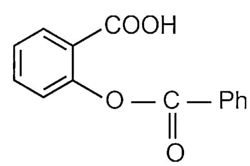
(2) 

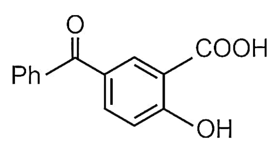
(3) 

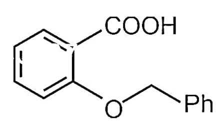
(4) 

94. When salicylic acid is treated with benzoyl chloride in presence of pyridine, the product formed is

(1) 

(2) 

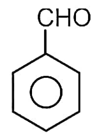
(3) 

(4) 

Space for Rough Work

95. An organic compound X on treatment with acidified $K_2Cr_2O_7$ gives compound Y which reacts with $I_2$ and sodium hydroxide to form tri-iodomethane. The compound X can be

(1) $CH_3OH$

(2) 

(3) $CH_3CHO$

(4) $CH_3CH(OH)CH_3$

96. Given below are two Statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

**Assertion (A):** Most of the carboxylic acids exist as dimer in vapour phase.

**Reason (R):** Carboxylic acids form intermolecular hydrogen bonds.

In the light of above statements, choose the most appropriate answer from the options given below.

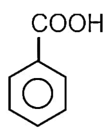
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

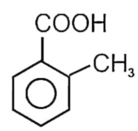
(2) (A) is correct but (R) is not correct

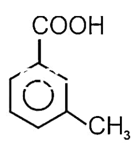
(3) (A) is not correct but (R) is correct

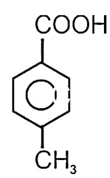
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

97. Which of the following is most acidic?

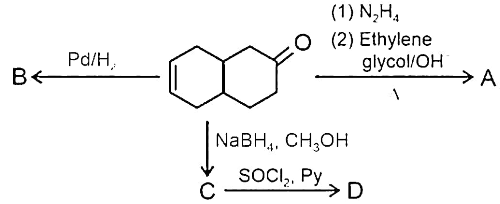
(1) 

(2) 

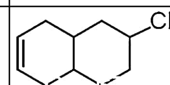
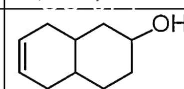
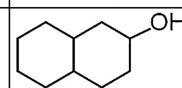
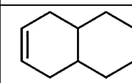
(3) 

(4) 

98. On the basis of following reactions




Match compound in column 'I' with their structure in column 'II' and choose the correct answer.

| | Column I | | Column II |
|---|---|---|---|
| (a) | A | (i) |  |
| (b) | B | (ii) |  |
| (c) | C | (iii) |  |
| (d) | D | (iv) |  |

(1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

99. Phenol can be distinguished from aliphatic alcohol with

(1) Tollen's reagent

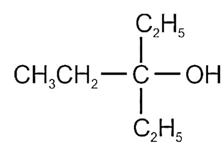
(2) Schiff's base

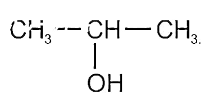
(3) Neutral $FeCl_3$

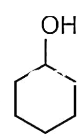
(4) HCl

100. Which alcohol on heating with Cu at 573 K gives alkene as major product?

(1) $CH_3CH_2CH_2OH$

(2) 

(3) 

(4) 

Space for Rough Work

# BOTANY

## SECTION - A

101. Verhulst-Pearl logistic growth is described by the equation

(1) $\frac{dN}{dt} = (b + d)N$ (2) $\frac{dN}{dt} = rN$

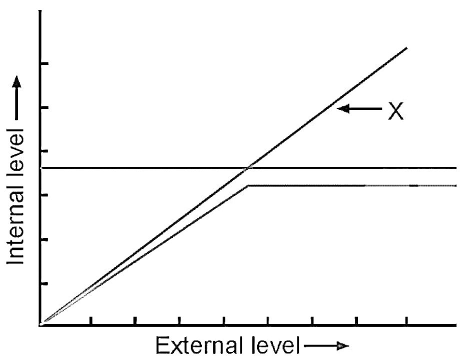
(3) $N_t = N_0 e^{rt}$ (4) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

102. To avoid summer-related problems some snails and fishes enter the stage of suspended development known as

(1) Hibernation (2) Aestivation

(3) Diapause (4) Desiccation

103. Consider the following diagrammatic representation of organismic response




Choose the **incorrect** statement for the 'X' in the above graph.

(1) These organisms change their body temperature with ambient temperature

(2) These organisms have mechanism to maintain constant internal temperature

(3) This type of response is shown by most of the animals and plants

(4) This type of response is shown by conformers.

104. Lichens are well known interaction between an alga and a fungus where alga has/shows

(1) Autotrophic nutrition and non-symbiotic relation with fungus

(2) Detrimental relation with fungus

(3) Parasitic relation with the fungus

(4) Symbiotic relation with the fungus

105. All of the following are attributes of a population, **except**

(1) Birth (2) Natality

(3) Mortality (4) Sex-ratio

106. Select the **odd** one out w.r.t. population interaction.

(1) Barnacles growing on the back of a whale

(2) Yellow-billed oxpecker sit on the back of black rhinoceros

(3) Epiphytes growing on mango plants

(4) Clown fish lives among sea anemone

107. In a lake, there are 30 *Eichhornia* plants last year and through reproduction, the current population have 45 plants. The birth rate will be

(1) 1 (2) 0.8

(3) 0.7 (4) 0.5

108. Read the following statements.

**Statement A:** Competition will occur only when resources present in the environment are limiting.

**Statement B:** Herbivores and plants appear to be more adversely affected by competition than carnivores.

In the light of above statements mark the **correct** option.

(1) Only statement A is correct

(2) Both statements A and B are correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

Space for Rough Work

109. Which among the following adaptations is **not** true for the parasites?
    (1) Low reproductive capacity
    (2) Loss of digestive system
    (3) Presence of adhesive organs
    (4) Loss of unnecessary sense organs

110. Which among the following is **not** considered as a role of predator in nature?
    (1) Transfer of energy across trophic levels
    (2) Maintain species diversity in a community
    (3) Complete eradication of prey population
    (4) Reduce the intensity of competition among prey species

111. When does the growth of a population following the exponential growth model is said to be zero?
    (1) When $\frac{N}{K}$ equals zero
    (2) When intrinsic rate of natural increase equals zero
    (3) When $\frac{N}{K}$ equals unity
    (4) When intrinsic rate of natural increase is exactly one

112. An ecologist studied the population of *Drosophila* in a given area. For a particular time period he found number of births 200, number of deaths 180, immigration 20 and emigration 30. Select the most appropriate impact w.r.t. the population size of *Drosophila*.
    (1) Expanding (2) Stable
    (3) Growth rate is zero (4) Declining

113. In which of the following biomes, the range of mean annual precipitation is minimum?
    (1) Coniferous forest
    (2) Tropical forest
    (3) Grassland
    (4) Temperate forest

114. Abiotic factors that lead to variation in the physical and chemical conditions of different habitats, include
    (a) Temperature (b) Light
    (c) Water (d) Soil
    The **correct** ones are
    (1) All (a), (b), (c) and (d)
    (2) Only (a) and (c)
    (3) Only (b) and (d)
    (4) Only (a), (b) and (c)

115. Select the **incorrect** statement from the following.
    (1) Pathogens are also a component of a habitat
    (2) Each organism has its specific niche and no other organism of the same species occupy the same niche
    (3) Some organisms can even survive in most harsh habitats such as permafrost polar regions
    (4) Trees in coniferous forests have needle like leaves

116. The salinity (parts per thousand) of sea water ranges between
    (1) 80 –100 (2) 1 – 5
    (3) 75 – 80 (4) 30 – 35

117. Eurythermal organisms include
    (1) Most of the mammals and birds
    (2) Most of the plants and animals
    (3) A very few mammals
    (4) Only birds

118. When we go at higher altitudes our body experiences altitude sickness. To overcome this conditions which of the following means are opted?
    (a) Increase the synthesis of haemoglobin
    (b) Increase binding capacity of haemoglobin with $O_2$
    (c) Increase breathing rate
    (1) (a) and (b) (2) (b) and (c)
    (3) (a) and (c) (4) All (a), (b) and (c)

**Space for Rough Work**

119. Cuckoo bird laying its eggs in crow's nest. This type of interaction is
(1) Mutualism (2) Predation
(3) Amensalism (4) Brood parasitism

120. Arrange the following levels of ecological organisations in ascending order of their complexity and choose the **correct** option.
a. Biological community
b. Biome
c. Ecosystem
d. Population
(1) c < d < a < b (2) d < c < a < b
(3) d < a < c < b (4) c < b < a < d

121. Biogas
(1) Does not include $H_2S$ as its constituent.
(2) Is produced in aeration tank during sewage treatment process.
(3) Is obtained by activity of only aerobic bacteria.
(4) Is produced in anaerobic sludge digester

122. Ganga and Yamuna action plan
(1) Was initiated by Indian council of agricultural research institute.
(2) Aims to save major rivers of our country from pollution.
(3) Was initiated in 1955.
(4) Focuses on use of waste water for domestic purpose.

123. The bioactive substance produced by a bacterium and used for patients who have undergone myocardial infarction is
(1) Streptokinase (2) Cyclosporin A
(3) Statins (4) Lactic acid

124. Conversion of milk to curd improves its nutritional value by increasing the amount of vitamin
(1) K (2) D
(3) A (4) $B_{12}$

125. During primary treatment in sewage treatment plants, the solids settle to form the __A__ and the supernatant forms the __B__.
Here A and B are

| | A | B |
|---|---|---|
| (1) | Primary effluent | Primary sludge |
| (2) | Primary sludge | Primary effluent |
| (3) | Activated sludge | Primary effluent |
| (4) | Primary sludge | Activated sludge |

126. Which of the following is true for biogas?
(1) Mixture of only inflammable gases
(2) Act as a fuel produced by algae
(3) Act as fuel produced by methanogens
(4) Can be used as a substrate for the formation of industrial products.

127. Methanobacterium is commonly found in all, **except**
(1) Excreta of herbivores like cow
(2) Aerobic sludge during sewage treatment
(3) Marshy areas
(4) Rumen of cattle

128. Match the following columns and select the **correct** option.

| | Column-I | | Column-II |
|---|---|---|---|
| a. | Ladybird | (i) | Most of them belongs to genus *Nucleopolyhedrovirus* |
| b. | *Bacillus thuringiensis* | (ii) | Free living fungi common in root ecosystem |
| c. | Baculovirus | (iii) | Useful in controlling aphids |
| d. | *Trichoderma* | (iv) | Introduced to control butterfly caterpillar |

(1) a(iii), b(iv), c(i), d(ii) (2) a(i), b(iii), c(iv), d(ii)
(3) a(i), b(ii), c(iii), d(iv) (4) a(iii), b(iv), c(ii), d(i)

Space for Rough Work

129. Which of the following option has only autotrophic organisms that can fix atmospheric nitrogen?
(1) *Anabaena, Aulosira* and *Rhizobium*
(2) *Nostoc*, *Glomus* and *Azotobacter*
(3) *Anabaena*, *Nostoc* and *Oscillatoria*
(4) *Azotobacter*, *Rhizobium,* Methanogens

130. Read the following statements.
**Assertion (A):** Plants having mycorrhizal association show resistance to root borne pathogens and tolerance to salinity.
**Reason (R) :** The algal symbiont absorbs phosphorus from soil and passes it to plant.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

131. Beverages like __A__ and __B__ are produced without distillation.
Choose the **correct** option to fill the blanks A and B.
(1) Rum and whisky (2) Beer and brandy
(3) Wine and beer (4) Whisky and brandy

132. The first antibiotic was discovered by Alexander Fleming. The source for this antibiotic was
(1) *Staphylococcus* (2) *Penicillium notatum*
(3) *Aspergillus niger* (4) *Acetobacter aceti*

133. Select the **incorrect** statement about biocontrol agents.
(1) They do not completely eradicate the pest and pathogens
(2) The use of biocontrol measures reduce our dependence on toxic chemical and pesticides
(3) These are toxic and extremely harmful to humans as they have been polluting our environment
(4) It refers to the use of biological methods for controlling plant disease and pests

134. Toddy is made by fermenting sap from
(1) Palm (2) Rice
(3) Wheat (4) Maize

135. Keoladeo national park is situated at __A__ and is famous for hosting __B__ in winter. Choose the **correct** option for A and B.

| | A | B |
|---|---|---|
| (1) | Rajasthan | Lion |
| (2) | Hazaribagh | Sparrow |
| (3) | Ranthambore | Tiger |
| (4) | Bharatpur | Siberian cranes |

**SECTION - B**

136. Which of the given household product(s) is produced by the bacterial fermentation?
(1) Bread
(2) Dosa and idli
(3) Toddy
(4) Wine

137. Mark the statement which is **not** true for antibiotics.
(1) It means 'pro life' in context of disease causing organism
(2) It is regarded as one of the most significant discoveries of the twentieth century
(3) These have greatly improved our capacity to treat deadly disease such as plague
(4) It was extensively used to treat American soldiers wounded in World War II

**Space for Rough Work**

138. The bioactive molecule that is competitive inhibitor of HMG CoA reductase is
(a) Produced by fungi
(b) Commercialised as blood cholesterol lowering agent
(c) Used as an immunosuppressive agent
(d) Helpful in removing oily stains from laundry
The **correct** ones are
(1) Only (a) and (b) (2) Only (c) and (d)
(3) Only (a), (b) and (c) (4) Only (b), (c) and (d)

139. Butyric acid is
(1) Used for commercial production of ethanol
(2) Produced by bacterium
(3) Used mainly as antibiotic
(4) Obtained from algae

140. Secondary treatment of STP includes
(A) Treatment of primary effluent with aerobic bacteria
(B) Removal of grit
(C) Sedimentation of flocs
(1) Only (A) (2) (A) and (B)
(3) (A) and (C) (4) Only (B)

141. Flocs are masses of
(1) Aerobic bacteria associated with fungal filaments
(2) Anaerobic bacteria associated with algal filaments
(3) Algae and fungi
(4) Cyanobacteria and viruses

142. Read the given statements and select the **correct** option.
**Statement A:** The greater the BOD of waste water, less is its polluting potential.
**Statement B:** BOD is a measure of organic matter present in the water.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

143. In logistic growth, a population growing in a habitat with a limited resources initially shows a __A__ followed by phase of __B__ then __C__ and finally are __D__.
Select the **correct** option and make the sentence of correct sense.

| | A | B | C | D |
|---|---|---|---|---|
| (1) | Lag phase | Acceleration | Deceleration | Asymptote |
| (2) | Asymptote | Deceleration | Acceleration | Lag phase |
| (3) | Lag phase | Acceleration | Asymptote | Deceleration |
| (4) | Deceleration | Asymptote | Lag phase | Acceleration |

144. Kangaroo rats in North American deserts is capable of meeting all of its water requirements
(1) By consuming the flesh of other animals
(2) By absorbing the dew present on the plants during night
(3) Through its internal fat oxidation
(4) From fleshy leaf of desert plant

145. Which of the pair of factors will decrease the population density of an area?
(1) Natality and immigration
(2) Mortality and Natality
(3) Immigration and emigration
(4) Mortality and emigration

146. The 'r' in the equation $\frac{dN}{dt} = rN$ is
(1) 0.12 for human population in 1981
(2) Intrinsic rate of natural increase
(3) Used to assess impact of only biotic factors on population growth
(4) Population density at time zero

Space for Rough Work

147. Graphical representation of proportion of various age groups of a population is

(1) Sex ratio (2) Percent cover

(3) Age pyramid (4) Biotic community

148. In which of the following population interactions, both interacting species are negatively affected?

(1) Commensalism (2) Competition

(3) Mutualism (4) Protocooperation

149. Exponential growth

(1) Occurs when resources are limiting in environment

(2) Is the realistic growth model found in nature

(3) Is not influenced by environmental factors

(4) Results in J-shaped curve

150. Read the following statements.

**Assertion (A):** In the polar sea, the aquatic mammals like seal have thick layer of fat which helps in reducing loss of body heat.

**Reason (R) :** Mammals from the colder climates generally have shorter ears and limbs to minimise heat loss.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

## ZOOLOGY

### SECTION - A

151. Select the **incorrect** statement w.r.t. AIDS.

(1) It was first reported in the year 1981.

(2) It is caused by Human Immuno-deficiency Virus.

(3) Transmission of HIV infection occurs only by sexual contact with an infected person.

(4) HIV does not spread by mere touch or physical contact.

152. Which of the following techniques/methods is not used in detection of cancers of the internal organs?

(1) Radiography

(2) Computed tomography

(3) Magnetic resonance imaging

(4) Radiotherapy

153. Complete the analogy by selecting the **correct** option.

________ : *Wuchereria* : : *Aedes* : Dengue virus

(1) *Anopheles* (2) *Aedes*

(3) *Culex* (4) *Plasmodium*

154. Heat and moisture help *Microsporum* to grow in the groin or between toes of an individual and causes a disease that is generally acquired from/by

(1) Soil

(2) Towel of healthy person

(3) Mosquito bite

(4) Viral spreading

155. The most serious form of malaria *i.e.*, malignant malaria which can be even fatal is caused by

(1) *P. vivax* (2) *P. falciparum*

(3) *P. malariae* (4) *P. ovale*

Space for Rough Work

156. One of the most infectious human ailment – the common cold generally occurs through
(1) Ingestion of contaminated food
(2) Bite of an infected mosquito
(3) Droplet inhalation from cough and sneezes
(4) Hyperactivity of immune system

157. Who among the following disproved the 'good humor' hypothesis of health?
(1) Hippocrates (2) William Harvey
(3) M.S. Swaminathan (4) Herbert Boyer

158. All of the following are important to maintain good health, **except**
(1) Balanced diet (2) Personal hygiene
(3) Sedentary habits (4) Yoga

159. Which vector is responsible for reported cases of chikungunya in many parts of India?
(1) *Gambusia*
(2) *Periplaneta americana*
(3) *Aedes* mosquito
(4) Tse-tse fly

160. The fertilisation of gametes of *Plasmodium* takes place in
(1) Mosquito's gut
(2) Human liver
(3) Human RBCs
(4) Mosquito's salivary gland

161. How many of the structures given in the box below are secondary lymphoid organs?

| |
|---|
| Spleen, Lymph nodes, Bone marrow, Thymus, Tonsils, Peyer's patches, Appendix |

Choose the **correct** option.
(1) Three (2) Four
(3) Five (4) Six

162. Consider the following statements and choose the most appropriate option.
**Statement (A):** All organisms belonging to protozoa and helminth can cause diseases in man and are called pathogens.
**Statement (B):** Most parasites are pathogens as they cause harm to the host by living in/on them.
(1) Only statement (A) is true
(2) Both statements (A) and (B) are true
(3) Both statements (A) and (B) are false
(4) Only statement (A) is false

163. Observe the diseases listed below.
(a) AIDS (b) Cancer
(c) Pneumonia (d) Common cold
How many of the above listed diseases is/are transmitted from one person to another?
(1) Four (2) Three
(3) One (4) Two

164. Read the following statements carefully.
(A) *Salmonella typhi* is a pathogenic bacterium that causes typhoid fever in human beings.
(B) The pathogens of typhoid generally enter the colon of small intestine through contaminated food and water.
(C) Typhoid fever can be confirmed by Widal test.
Select the most appropriate option.
(1) Statement (A) is true while statements (B) and (C) are false.
(2) Statements (A) and (B) are true while statement (C) is false.
(3) Statements (A) and (C) are true while statement (B) is false.
(4) All statements (A), (B) and (C) are true.

Space for Rough Work

165. An immunoglobulin molecule consists all of the following components, **except**
   (1) Light chains (2) Heavy chains
   (3) Disulphide bonds (4) Antibody binding site
166. Which type of antibodies primarily increase in response to allergens like dust, animal dander, *etc*?
   (1) IgA (2) IgE
   (3) IgM (4) IgG
167. Select an opiate drug which suppresses brain function, relieves intense pain and thus is used as an analgesic, post surgery.
   (1) Cocaine (2) Marijuana
   (3) Morphine (4) Hashish
168. Mucosa Associated Lymphoid Tissue (MALT) constitutes about __X__ per cent of the lymphoid tissue in human body. Here 'X' is
   (1) 50 (2) 60
   (3) 70 (4) 80
169. **Assertion (A):** During malarial infection, the chill and high fever is usually observed at regular intervals.
   **Reason (R):** A toxic substance, haemozoin is released by the rupture of RBCs in each cycle.
   In the light of above statements, choose the **correct** option.
   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
   (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
   (3) (A) is true but (R) is false
   (4) Both (A) and (R) are false
170. Read the statements (P) and (Q) given below and select the **correct** option.
   **Statement (P) :** *Haemophilus influenzae* and *Trichophyton* cause common cold in human beings.
   **Statement (Q) :** Common cold is characterised by nasal congestion and discharge, sore throat, cough, headache, *etc*.
   (1) Both statements (P) and (Q) are true
   (2) Both statements (P) and (Q) are false
   (3) Statement (P) is true but statement (Q) is false
   (4) Only statement (Q) is true
171. A large bean-shaped organ which has a large reservoir of erythrocytes is
   (1) Liver (2) Spleen
   (3) Pancreas (4) Thymus
172. The chronic use of drugs and alcohol damages liver and cause
   (1) Bronchitis (2) Cirrhosis
   (3) Emphysema (4) AIDS
173. In a monomeric antibody molecule, the total number of intra-chain disulphide bonds is equal to
   (1) Six (2) Eight
   (3) Twelve (4) Four
174. Observe the figure given below.

   Identify the above given figure and select the **correct** option w.r.t. it.
   (1) Extracts of this plant are generally taken by oral ingestion
   (2) Extracts of this plant belong to cannabinoids
   (3) This plant is native to South America
   (4) Morphine is extracted from the latex of this plant

Space for Rough Work

175. Observe the facts/features listed below

(a) White and odourless

(b) Bitter and crystalline

(c) Slows down body function

(d) Taken by snorting and injection

Above mentioned facts/features are associated with the chemical whose receptors are present in

(1) Heart only

(2) Brain only

(3) Gastrointestinal tract only

(4) Both CNS and GIT

176. Which of the following substances is given to cancer patients that activates their immune system and helps in destroying the tumor?

(1) Smack (2) $\alpha$-interferons

(3) IgE (4) Histamine

177. All of the following are symptoms associated with pneumonia, **except**

(1) Gray to bluish discoloration of lips

(2) Cough and headache

(3) Diarrhoea

(4) Fever

178. The period between ________ years of age is termed as adolescence period.

Choose the option which fills the blank **correctly**.

(1) 10 – 12 (2) 8 – 12

(3) 12 – 18 (4) 22 – 30

179. Vaccine produced using genetic engineering or recombinant DNA technique is/has

(1) Eradicated polio

(2) Lower availability for immunisation

(3) Hepatitis B vaccine

(4) Used for passive immunisation

180. Choose the **incorrect** statement.

(1) Tobacco contains a large number of harmful chemical substances including nicotine, an alkaloid.

(2) Nicotine stimulates adrenal gland to release adrenaline and noradrenaline.

(3) Smoking decreases carbon monoxide (CO) content in blood and increases concentration of haem-bound oxygen.

(4) Smoking is associated with the increased incidences of emphysema and gastric ulcer.

181. The *Plasmodium* enters the mosquito's body as

(1) Sporozoites (2) Gametocytes

(3) Zygotes (4) Gametes

182. Which of the following structures of our immune system serve to trap the blood-borne micro-organisms and also act as graveyard of RBCs?

(1) Thymus (2) Bone-marrow

(3) Appendix (4) Spleen

183. Mucus coating of the epithelium lining the urogenital tract is included under the category of

(1) Physiological barriers of innate immunity

(2) Physical barriers of innate immunity

(3) Cellular barriers of innate immunity

(4) Passive acquired immunity

184. Which of the following is **not** true w.r.t. acquired immunity?

(1) It is pathogen specific and is characterised by memory

(2) When our body encounters a pathogen for the first time, it produces anamnestic response.

(3) Primary immune response is of low intensity.

(4) Subsequent encounter with the same pathogen elicits a highly intensified secondary response.

**Space for Rough Work**

185. Match Column I with Column II and choose the **correct** option.

| | Column I | | Column II |
|---|---|---|---|
| a. | Antibodies | (i) | Cell-mediated immunity |
| b. | T-lymphocytes | (ii) | Humoral immunity |
| c. | Monocytes | (iii) | Phagocytosis |

(1) a(i), b(iii), c(ii) (2) a(iii), b(ii), c(i)
(3) a(iii), b(i), c(ii) (4) a(ii), b(i), c(iii)

**SECTION - B**

186. All of the following are few common causes, which motivate youngsters towards drug and alcohol use, **except**
(1) Need for adventure and excitement
(2) Experimentation
(3) Education and counselling
(4) Curiosity

187. Choose the **incorrect** option w.r.t. filariasis in humans.
(1) It is caused by a helminth, *W. malayi*.
(2) The lymphatic vessels of the lower limbs are usually affected.
(3) Characterised as a rapid developing chronic inflammation of the organs in which the pathogens do not survive more than a week.
(4) The genital organs also get affected

188. Internal bleeding, muscular pain, anemia and blockage of intestinal passage are symptoms of a helminthic disease caused by
(1) *Epidermophyton* (2) *Ascaris*
(3) *Wuchereria* (4) *Plasmodium*

189. Read the following carefully
(a) Drop in academic performance
(b) Change in sleeping and eating habits
(c) Increased interest in personal hygiene
(d) Better relations with family and friends
How many of the above is/are **not** true w.r.t. the common warning signs of drug abuse among youth?
(1) Three (2) Four
(3) Two (4) Zero

190. Lysozyme present in saliva destroys certain types of
(1) Bacteria (2) Fungi
(3) Viruses (4) Nematodes

191. **Assertion (A):** Virus infected cells secrete a polysaccharide that is a cytokine barrier of innate immunity.
**Reason (R):** Interferons keep a check on the spreading of virus to non-infected cells.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

192. Consider the symptoms listed below
(a) Constipation
(b) Abdominal pain and cramps
(c) Stools with excess mucous and blood
Identify the disease and select the **correct** option w.r.t. it.
(1) This disease is caused by a bacterium, *E. histolytica.*
(2) Housefly acts as mechanical carrier to transmit parasite of this disease.
(3) The pathogen of this disease is a parasite in the small intestine of human.
(4) It is usually transmitted *via* nasal droplets.

193. All of the following are correct w.r.t. cannabinoids, **except**

(1) Obtained from the inflorescences of the plant *Cannabis sativa*

(2) Generally taken by inhalation and oral ingestion

(3) Known for their inhibitory effects on transport of dopamine

(4) Cannabinoid receptors are present principally in the brain.

194. Mary Mallon, who was a cook by profession was known to be a carrier of a disease named

(1) Amoebiasis (2) Typhoid

(3) Filariasis (4) Ascariasis

195. Select the **incorrect** match among the following.

| | | | |
|---|---|---|---|
| (1) | Colostrum | – | Passive immunity |
| (2) | $\alpha$-interferon | – | Biological response modifier |
| (3) | Mast cells | – | Histamine, serotonin, adrenaline |
| (4) | CT scan | – | Uses X-rays |

196. Which of the following is the part of innate immunity?

(1) RBCs (2) B-lymphocytes

(3) NK cells (4) Platelets

197. Barbiturates are normally used as medicine to help patients cope with

(1) AIDS (2) Malaria

(3) Insomnia (4) Ascariasis

198. The immune response that is mainly responsible for the rejection of kidney graft is

(1) Humoral immune response

(2) Cell mediated immune response

(3) Auto-immune response

(4) Innate immune response

199. Choose the **correct** statement w.r.t. addiction and dependence.

(1) Addiction is a physical attachment to certain effects associated with drugs and alcohol.

(2) With the repeated use of drugs, the tolerance level of the receptors present in our body decreases.

(3) Withdrawal syndrome is characterised by anxiety, shakiness, nausea and sweating.

(4) Dependence leads the patient to obey all social norms in order to get sufficient funds to satiate his/her needs.

200. The side-effects of the use of anabolic steroids in human females do not include

(1) Masculinisation

(2) Enlargement of clitoris

(3) Breast enlargement

(4) Depression



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